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Module 6: Genetics, Evolution and Ecosystems

OCR A A-Level Biology (H420)

6.1 Cellular Control

Gene Expression and the Genetic Code

A gene is a sequence of DNA nucleotides that codes for a polypeptide (or functional RNA). The genetic code is a triplet code: each sequence of three bases (codon) specifies one amino acid. The code is degenerate (most amino acids are coded by more than one codon), non-overlapping (each base belongs to only one codon), and universal (the same codons specify the same amino acids in almost all organisms, providing evidence for common ancestry).

Transcription

Transcription produces a single-stranded mRNA molecule complementary to the template (antisense) strand of DNA. RNA polymerase binds to the promoter region upstream of the gene, unwinds and unzips a short section of DNA, and synthesises mRNA by adding complementary RNA nucleotides in the 5' to 3' direction (using uracil instead of thymine). In eukaryotes, the primary mRNA transcript (pre-mRNA) contains introns (non-coding sequences) that are removed by splicing before the mature mRNA leaves the nucleus through nuclear pores.

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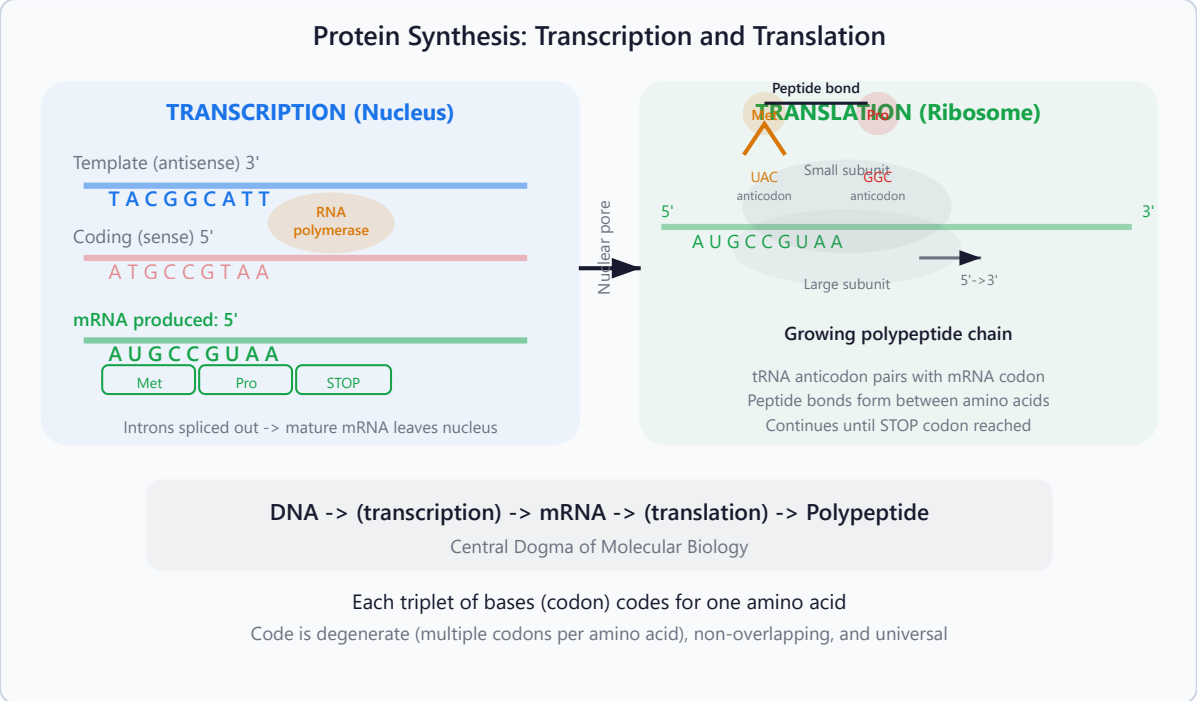
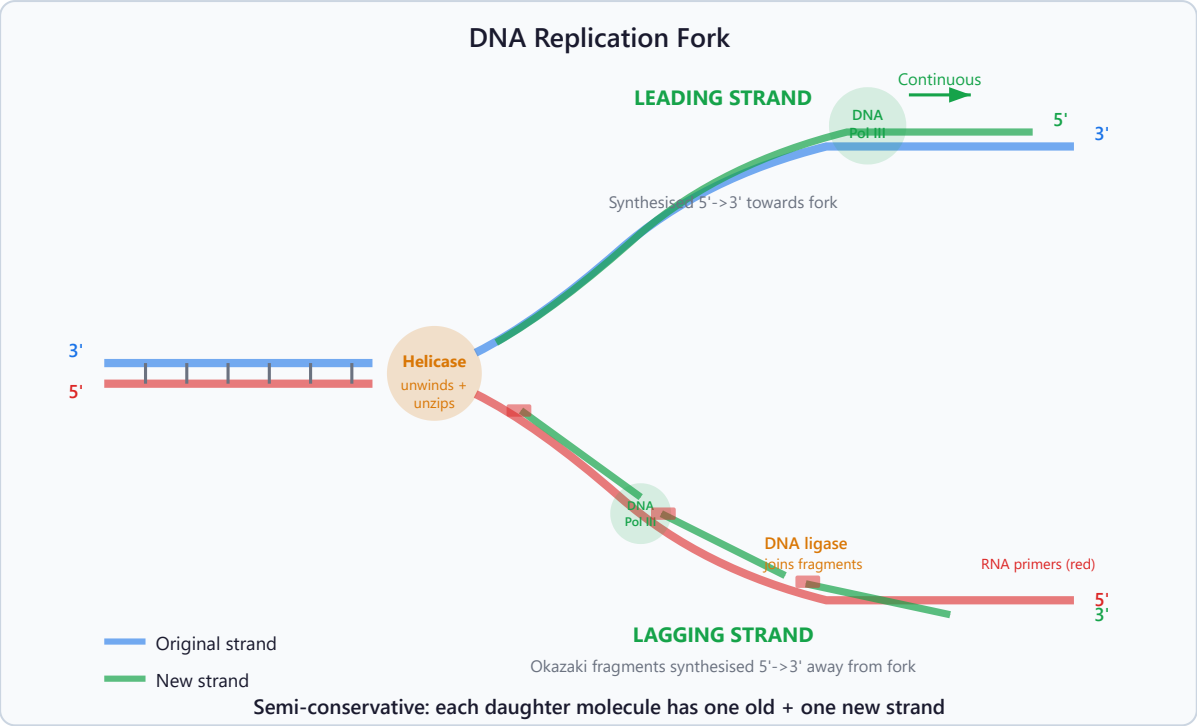


Translation

Translation occurs at ribosomes in the cytoplasm. The mRNA binds to a ribosome, and transfer RNA (tRNA) molecules bring specific amino acids to the ribosome. Each tRNA has an anticodon complementary to a specific mRNA codon and carries the corresponding amino acid. The ribosome moves along the mRNA, reading codons in sequence. At each codon, a complementary tRNA binds, and a peptide bond is formed between adjacent amino acids by the ribosome (peptidyl transferase activity). Translation continues until a stop codon is reached, at which point the polypeptide is released.



Gene Mutations

Mutations are changes in the base sequence of DNA. Point mutations involve a single nucleotide:

- **Substitution** — one base replaced by another. May be silent (codes for same amino acid due to degenerate code), missense (codes for different amino acid, may or may not affect protein function), or nonsense (creates a premature stop codon, producing a truncated non-functional protein)
- **Insertion** — an extra base added to the sequence, causing a frameshift (all codons downstream are altered, producing a completely different amino acid sequence)
- **Deletion** — a base removed from the sequence, also causing a frameshift mutation with severe consequences for protein structure and function

Mutations arise spontaneously during DNA replication (errors by DNA polymerase) or are induced by mutagens (UV radiation causes thymine dimers, ionising radiation causes strand breaks, certain chemicals intercalate between bases or act as base analogues). Most mutations are neutral or harmful, but occasionally a mutation produces a beneficial change that increases fitness and may be selected for in the population.

Regulation of Gene Expression

Not all genes are expressed in all cells at all times. Gene expression is regulated at multiple levels:

- **Transcriptional control** — transcription factors (activators and repressors) bind to regulatory regions of DNA, promoting or inhibiting RNA polymerase binding. Epigenetic modifications (DNA methylation silences genes; histone acetylation opens chromatin, enabling transcription) provide heritable changes in expression without altering the DNA sequence.
- **Post-transcriptional control** — alternative splicing allows different mRNA variants (and therefore different proteins) to be produced from the same gene by including different combinations of exons. mRNA degradation rate affects protein production levels.
- **Translational control** — regulatory molecules can inhibit translation initiation or affect ribosome binding.
- **Post-translational control** — proteins may require modification (glycosylation, phosphorylation, cleavage of signal sequences) to become functional; protein degradation by proteasomes controls protein lifespan.

Apoptosis

> **Apoptosis:** Programmed cell death — a controlled, genetically regulated process in which cells are systematically dismantled without causing inflammation or damage to surrounding tissue. The cell shrinks, the chromatin condenses, the DNA is fragmented by endonucleases, the cell breaks into apoptotic bodies, and these are phagocytosed by neighbouring cells or macrophages.

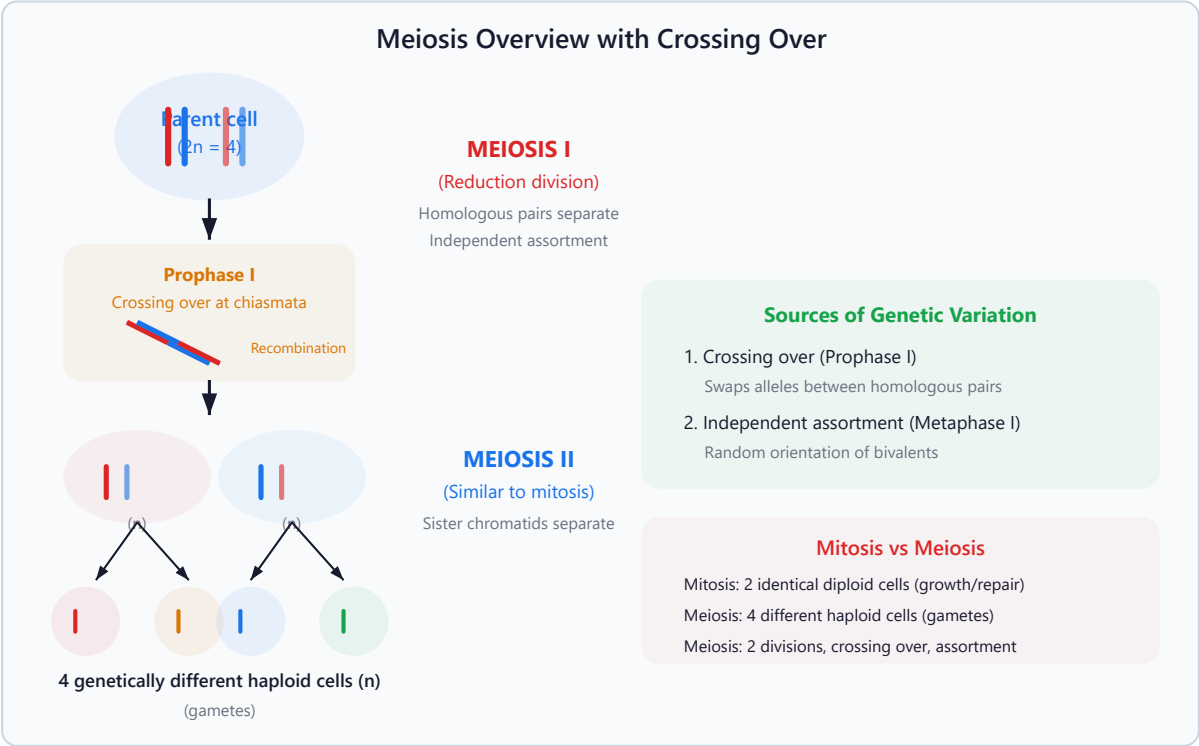
Apoptosis is essential for normal development (removing webbing between digits in embryos, sculpting neural connections), maintaining tissue homeostasis (balancing cell division with cell death), and immune function (destroying virus-infected cells and self-reactive lymphocytes). Failure of apoptosis can contribute to cancer (cells that should die continue to proliferate). Excessive apoptosis contributes to neurodegenerative diseases.

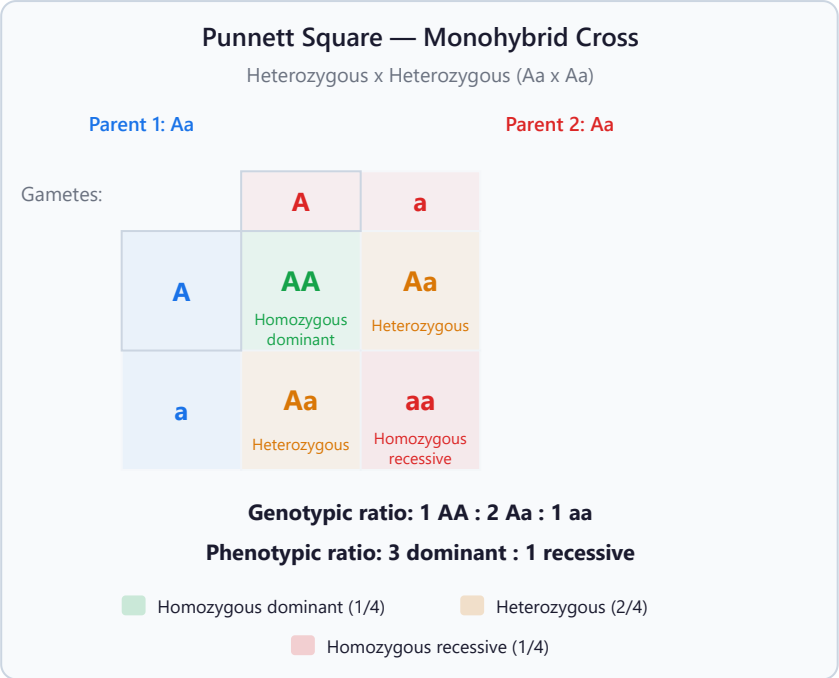
6.2 Patterns of Inheritance

Key Genetic Terms

Term	Definition

Gene	A sequence of DNA that codes for a polypeptide or functional RNA
Allele	An alternative version of a gene, occupying the same locus on homologous chromosomes
Locus	The specific position of a gene on a chromosome
Genotype	The combination of alleles an organism possesses for a particular gene
Phenotype	The observable characteristics of an organism, resulting from genotype and environmental interactions
Homozygous	Having two identical alleles for a particular gene (e.g., AA or aa)
Heterozygous	Having two different alleles for a particular gene (e.g., Aa)
Dominant	An allele that is expressed in both homozygous and heterozygous conditions
Recessive	An allele that is only expressed in the homozygous condition
Codominant	Both alleles are fully expressed in the heterozygote (e.g., blood group AB)





Monohybrid Inheritance

Monohybrid crosses involve a single gene with two alleles. A cross between two heterozygotes (Aa x Aa) produces offspring in the expected ratio of 3 dominant phenotype : 1 recessive phenotype (genotypic ratio 1 AA : 2 Aa : 1 aa). A test cross (crossing an individual of dominant phenotype with a homozygous recessive) determines whether the dominant individual is homozygous (all offspring show dominant phenotype) or heterozygous (approximately 1:1 ratio of dominant to recessive).

Dihybrid Inheritance

Dihybrid crosses involve two genes on different chromosomes (unlinked genes that assort independently according to Mendel's second law). A cross between two double heterozygotes (AaBb x AaBb) produces offspring in the expected ratio of 9:3:3:1 (9 both dominant : 3 first dominant only : 3 second dominant only : 1 both recessive). Deviations from this ratio may indicate linkage, epistasis, or lethal alleles.

Sex Linkage

Genes located on the X chromosome show sex-linked inheritance because males (XY) have only one X chromosome and will express any allele present on it (hemizygous), while females (XX) can be heterozygous carriers. X-linked recessive conditions (e.g., haemophilia, red-green colour blindness) are more common in males because they only need one copy of the recessive allele to be affected, whereas females need two copies.

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Epistasis

Epistasis occurs when one gene masks or modifies the expression of another gene at a different locus. In **recessive epistasis**, the homozygous recessive genotype at the epistatic locus prevents expression of the other gene, modifying the 9:3:3:1 ratio to 9:3:4 (e.g., coat colour in Labrador retrievers where ee genotype produces yellow regardless of the B gene). In **dominant epistasis**, the presence of a dominant allele at the epistatic locus masks expression of the other gene, producing a 12:3:1 ratio.

Chi-Squared Test

The chi-squared test determines whether observed results differ significantly from expected results (i.e., whether deviations from expected ratios are due to chance or indicate that the genetic model is incorrect).

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Chi-squared = sum of [(O - E)^2 / E]

Where O = observed frequency, E = expected frequency
Degrees of freedom = number of categories - 1

If calculated chi-squared > critical value at p = 0.05:
  Reject null hypothesis (significant difference exists)
If calculated chi-squared < critical value at p = 0.05:
  Accept null hypothesis (no significant difference, results fit expected ratio)
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Exam Tip: When performing genetic crosses in exams, always show: a clear genetic diagram with parental phenotypes and genotypes, gametes in circles, a Punnett square, offspring genotypes, offspring phenotypes, and the expected ratio. For chi-squared, show all working in a table format and state your conclusion clearly with reference to the critical value and degrees of freedom.

6.3 Manipulating Genomes

DNA Technology

Restriction endonucleases are bacterial enzymes that cut DNA at specific recognition sequences (usually palindromic sequences of 4-8 bases). They produce either blunt ends or sticky ends (short single-stranded overhangs). Sticky ends are particularly useful because they can base-pair with complementary sticky ends from other DNA fragments cut with the same enzyme, facilitating gene insertion.

DNA ligase joins DNA fragments by catalysing the formation of phosphodiester bonds between adjacent nucleotides, permanently sealing the sugar-phosphate backbone. This is used to insert genes into vectors and to join Okazaki fragments during replication.

Polymerase Chain Reaction (PCR)

PCR amplifies specific DNA sequences in vitro, producing millions of copies from a tiny sample. The three-step cycle is repeated 25-35 times:

1. **Denaturation (95 degrees Celsius)** — hydrogen bonds between complementary bases are broken, separating the double helix into two single strands
2. **Annealing (50-65 degrees Celsius)** — short DNA primers (complementary to sequences flanking the target region) bind to the template strands, providing a starting point for DNA synthesis
3. **Extension (72 degrees Celsius)** — heat-stable Taq DNA polymerase (from Thermus aquaticus) adds complementary nucleotides to the 3' end of each primer, synthesising new DNA strands

Each cycle doubles the number of copies, so after n cycles there are 2^n copies of the target sequence. PCR is used in forensic DNA profiling, genetic testing, diagnosis of infections, and preparation of DNA for sequencing or cloning.

Gel Electrophoresis

Gel electrophoresis separates DNA fragments by size. DNA samples (often cut with restriction enzymes) are loaded into wells in an agarose gel and an electric field is applied. DNA fragments move towards the positive electrode (anode) because DNA is negatively charged (phosphate groups). Smaller fragments move faster and further through the gel matrix. The separated fragments are visualised using DNA stains or radioactive/fluorescent probes.

Applications include DNA profiling (comparing fragment patterns for forensic identification or paternity testing), restriction mapping, checking PCR products, and analysing results of genetic

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engineering.

Genetic Engineering

Genetic engineering involves isolating a gene of interest and inserting it into a different organism (creating a transgenic or genetically modified organism). The general steps are:

1. **Gene isolation** — using restriction enzymes (from genomic DNA), reverse transcriptase (from mRNA to create cDNA), or chemical synthesis
2. **Gene insertion into a vector** — the gene is ligated into a vector (plasmid, bacteriophage, or viral vector) that has been cut with the same restriction enzyme. The vector contains an origin of replication and selectable markers (antibiotic resistance genes or reporter genes).
3. **Transformation** — the recombinant vector is introduced into host cells (e.g., using heat shock, electroporation, or calcium chloride treatment for bacteria; gene guns or Agrobacterium for plants)
4. **Selection** — successfully transformed cells are identified using selectable markers (e.g., growth on antibiotic plates, fluorescence under UV)
5. **Expression** — the transgene is transcribed and translated in the host cell to produce the desired protein

Examples: insulin production in bacteria (human insulin gene in E. coli), golden rice (beta-carotene biosynthesis genes from daffodil and bacteria inserted into rice), Bt crops (Bacillus thuringiensis toxin gene in corn/cotton for insect resistance).

Gene Therapy

Gene therapy aims to treat genetic disorders by introducing functional copies of defective genes into patient cells. **Somatic gene therapy** targets body cells and is not inherited (e.g., treating cystic fibrosis by delivering functional CFTR genes to lung epithelial cells via viral vectors or liposomes). **Germ line gene therapy** modifies gametes or early embryos and would be inherited by offspring; it is currently prohibited in humans due to ethical concerns about unknown long-term effects and consent issues.

Challenges include: difficulty delivering genes to sufficient target cells, ensuring long-term expression (transgenes may be silenced by epigenetic mechanisms), immune responses to viral vectors (limiting repeated treatments), risk of insertional mutagenesis (gene inserting into or near a tumour

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suppressor or proto-oncogene, potentially causing cancer), and short duration of effect requiring repeated treatments as target cells are replaced.

CRISPR-Cas9 genome editing: A newer approach that allows precise editing of specific DNA sequences rather than just adding a functional gene copy. The Cas9 enzyme (guided by a synthetic guide RNA) cuts DNA at a precise location, and the cell's repair mechanisms can then delete, correct, or insert sequences. CRISPR is more precise than traditional gene therapy, can target any gene, and can correct mutations at their source. However, off-target effects (cutting at unintended sites), mosaicism, and ethical concerns about germline editing remain significant challenges.

> **Exam Tip:** When discussing genetic engineering and gene therapy, always consider: the potential benefits (treating previously incurable diseases, improving crop yields, producing medicines), the risks (unknown long-term effects, ecological concerns for GMOs, ethical issues), and the regulatory frameworks in place. Exam questions often require a balanced evaluation of both sides.

Bioinformatics and Genome Sequencing

Bioinformatics uses computational tools to analyse biological data, particularly DNA and protein sequences. Genome sequencing determines the complete nucleotide sequence of an organism's DNA. Applications include identifying genes associated with disease, comparing genomes between species to determine evolutionary relationships, developing personalised medicine based on individual genetic profiles, and identifying drug targets.

6.4 Cloning and Biotechnology

Natural and Artificial Cloning in Plants

Plants can reproduce asexually through natural vegetative propagation (runners, tubers, bulbs), producing genetically identical offspring (clones). Artificial cloning methods include taking cuttings (stems placed in rooting hormone produce adventitious roots), tissue culture (micropropagation: small pieces of plant tissue grown on sterile nutrient medium with growth regulators produce callus tissue, which differentiates into plantlets), and grafting.

Micropropagation is valuable for: rapid production of large numbers of genetically identical plants, preserving rare species, producing disease-free stock from meristem tips (which are typically virus-free), and commercial propagation of desirable cultivars.

Cloning in Animals

Artificial animal cloning includes: splitting embryos at an early stage (each part develops into a genetically identical organism) and somatic cell nuclear transfer (SCNT), where the nucleus from a differentiated adult cell is transferred into an enucleated egg cell. The reconstructed egg is stimulated to divide and develops into an organism genetically identical to the nuclear donor (e.g., Dolly the sheep).

Arguments for cloning: preserving endangered species, producing transgenic animals for medicine, understanding cell differentiation. Arguments against: low success rates, health problems in clones (premature ageing, organ defects), reduced genetic diversity, ethical concerns about the welfare of surrogate mothers and cloned animals.

Biotechnology

Biotechnology uses living organisms or their products in industrial processes. Fermentation technology uses microorganisms in controlled conditions (bioreactors/fermenters) to produce useful products: antibiotics (penicillin from Penicillium), enzymes (proteases, lipases for biological washing powders), food products (yoghurt, cheese, bread, beer), biofuels (ethanol from yeast fermentation of sugarcane/corn), and recombinant proteins (insulin, growth hormone from genetically modified bacteria).

Bioreactor conditions are carefully controlled: temperature (thermostat and cooling jacket), pH (buffers or acid/alkali addition), oxygen supply (sterile air sparging and impeller mixing), nutrient supply, and aseptic conditions (preventing contamination by unwanted microorganisms).

6.5 Ecosystems

Energy Transfer Through Ecosystems

> **Ecosystem:** All the living organisms (biotic community) and their non-living environment (abiotic factors) in a defined area, interacting as a system through nutrient cycling and energy

> flow.

Energy enters ecosystems as sunlight and is fixed by producers (autotrophs) through photosynthesis. Only approximately 1-3% of available light energy is converted to chemical energy in organic molecules. Energy is transferred through trophic levels (producer, primary consumer, secondary consumer, tertiary consumer) but significant losses occur at each transfer:

- Not all biomass at one trophic level is consumed (some dies and enters the decomposer food chain)
- Not all ingested material is assimilated (some is egested as faeces)
- Energy is lost as heat through cellular respiration at each trophic level

Typically only 10-20% of energy is transferred between trophic levels. This explains why food chains rarely exceed 4-5 trophic levels (insufficient energy remains to support further levels) and why biomass decreases at each successive trophic level (pyramids of biomass).

Net primary productivity (NPP) = Gross primary productivity (GPP) - Respiratory losses (R). NPP represents the energy available to consumers. Net production at consumer levels = energy assimilated - respiratory losses.

Nutrient Cycles

The Carbon Cycle: Carbon enters the biotic community when producers fix CO2 during photosynthesis into organic molecules. Carbon is transferred through food chains as organisms feed. Carbon is returned to the atmosphere as CO2 through respiration (all organisms), combustion of fossil fuels and biomass, and decomposition of dead organic matter by saprobiotic microorganisms. Carbon is stored long-term in fossil fuels (coal, oil, gas formed from ancient organic matter), in limestone (calcium carbonate from marine organisms), in ocean dissolved CO2, and in living biomass.

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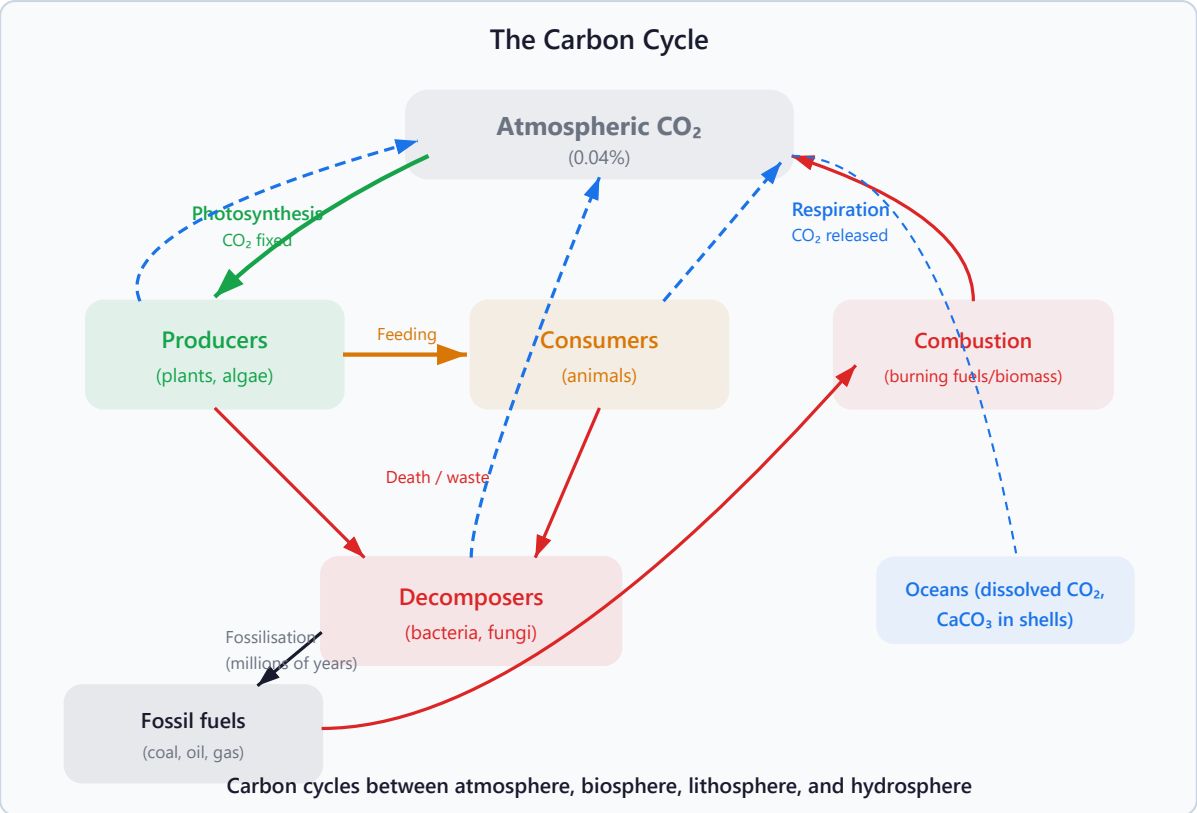


The Nitrogen Cycle: Atmospheric nitrogen (N2) is unavailable to most organisms. Key processes include:

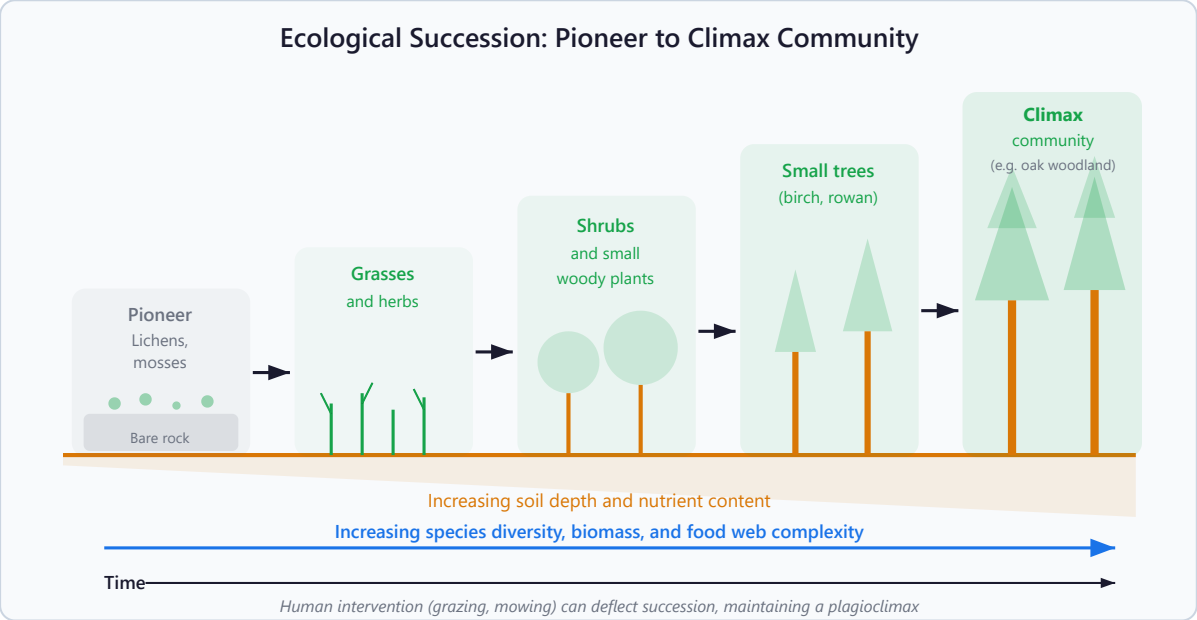
- **Nitrogen fixation** — conversion of N2 to ammonia (NH3) by nitrogen-fixing bacteria (Rhizobium in root nodules of legumes, or free-living Azotobacter), or by lightning
- **Nitrification** — oxidation of ammonia to nitrite (by Nitrosomonas) then to nitrate (by Nitrobacter) by nitrifying bacteria in aerobic soil
- **Assimilation** — plants absorb nitrate ions from soil and use them to synthesise amino acids and nucleotides; animals obtain nitrogen by consuming plant or animal protein
- **Ammonification** — decomposers break down organic nitrogen compounds in dead matter and excretory products, releasing ammonia
- **Denitrification** — anaerobic denitrifying bacteria (Pseudomonas) convert nitrate back to N2, returning it to the atmosphere (occurs in waterlogged, anaerobic soils)

Ecological Succession

Succession is the directional change in community composition over time. **Primary succession** occurs on previously uncolonised substrates (bare rock, volcanic lava, sand dunes) where pioneer species (lichens, mosses) colonise first, modifying the environment to make it suitable for subsequent

species. **Secondary succession** occurs where an existing community has been disturbed (fire, deforestation, abandoned farmland) but soil and seed bank remain.

During succession: species diversity increases, food webs become more complex, biomass accumulates, soil depth and nutrient content increase, and the community becomes more stable. The final stable community in equilibrium with the prevailing climatic conditions is the **climax community** (e.g., temperate deciduous woodland in the UK). Deflected succession occurs when human activity (grazing, mowing, burning) prevents the climax from developing, maintaining a plagioclimax community (e.g., grassland maintained by sheep grazing).



Conservation and Sustainability

Conservation aims to maintain biodiversity through protection and management of habitats and species. Reasons for conservation include: ecological (maintaining ecosystem stability and services such as pollination, water purification, climate regulation), economic (sustainable harvesting, ecotourism, genetic resources for crop improvement and medicine), ethical (intrinsic value of species, responsibility to future generations), and aesthetic (cultural and recreational value of natural environments).

Conservation strategies include: protected areas and nature reserves (in situ conservation), seed banks and zoos (ex situ conservation), habitat management (coppicing, controlled burning, grazing regimes to maintain diversity), legal protection (CITES, national wildlife legislation), reintroduction programmes, and sustainable resource management (fishing quotas, selective logging, replanting).

Sustainability means meeting current needs without compromising the ability of future generations to meet their own needs. In ecological terms, this means harvesting resources at a rate that allows populations to replenish themselves, maintaining ecosystem services, and minimising pollution and habitat destruction.

> **Exam Tip:** When discussing conservation, always link specific strategies to the threats they address and evaluate their effectiveness and limitations. For example, seed banks preserve genetic diversity but cannot conserve complex ecological relationships. Protected areas conserve habitats but may be too small to support viable populations of large predators without ecological corridors connecting them.

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